

REU Project Title: Trade-Offs Between Success-Attack Rate and Bandwidth Overhead for Timing-Based Webpage Fingerprinting Attacks

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Research Summary

During my REU, I worked on a project titled “Trade-Offs Between Success-Attack Rate and Bandwidth Overhead for Timing-Based Webpage Fingerprinting Attacks,” which studies how well the Tamaraw defense protects Tor users against website fingerprinting attacks. Tor encrypts traffic but still leaks timing, direction, and size patterns that allow powerful machine learning attacks to infer which web page a user visits. Tamaraw is a traffic-shaping defense that sends packets at fixed intervals and pads flows to more uniform lengths, with the goal of hiding these patterns. However, doing so introduces bandwidth overhead, and the practical trade-off between privacy and cost was not well understood across a broad range of configurations, especially under the stricter “one-page” evaluation setting.

My main research goal was to quantify how much extra bandwidth Tamaraw must add to significantly reduce attack success, and to understand whether those settings are realistic for the Tor network. To do this, I first reviewed prior work on Tor, website fingerprinting attacks, and Tamaraw and related defenses. Then I adapted an existing website fingerprinting pipeline to integrate Tamaraw as a configurable module. In our experiments, each traffic trace from a closed-world dataset of 95 monitored web pages was passed through Tamaraw, which padded packets using a Pad Length (PadL) parameter that directly controls bandwidth overhead. For each PadL value, I generated a one-page dataset and trained four strong website fingerprinting attacks—Deep Fingerprinting, Tik-Tok, Robust Fingerprinting, and Var-CNN—on the defended traces. For every

model and configuration, I measured standard metrics such as accuracy and F1 score, and matched them to the corresponding bandwidth overhead.

The results showed a clear privacy–performance trade-off. As PadL increased, bandwidth overhead rose from roughly 100% to 400%, meaning Tamaraw sometimes sent two to five times as many packets as the original Tor traffic. In parallel, attack accuracy and F1 score decreased, confirming that Tamaraw does weaken modern attacks. However, the drop in performance was gradual rather than abrupt: even at the highest overheads I tested, all four attacks still performed well above random guessing. This suggests that while Tamaraw can meaningfully reduce the effectiveness of powerful website fingerprinting attacks, it often needs very high overhead to do so, raising concerns about scalability and usability if deployed widely in the Tor network.

Beyond the technical findings, this project was my first real research experience, and it taught me what it actually means to do research in computer security. I learned how to read and synthesize research papers, work with existing codebases, design experiments, and interpret noisy results rather than just run “one-off” tests. I also gained experience presenting my work to broader audiences through a research poster, which I presented at the UTA College of Engineering Innovation Day and at the 2026 Discover Symposium. Working closely with Dr. Rémi Chou and Dr. Vamoua Yachongka, I developed stronger skills in experimental design, data analysis, and technical communication, and this experience confirmed my interest in continuing research on privacy, security, and networked systems.